

ODTE Asset Pricing

Almeida, Freire, Hizmeri

Jean-Sébastien Fontaine—Discussant

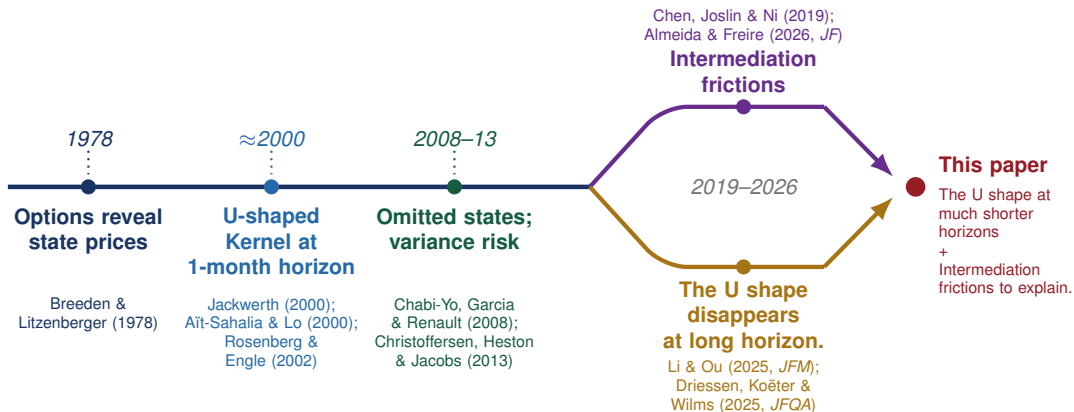
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The opinions expressed here are mine only and do not necessarily reflect the views of the Bank of Canada.

Motivation

- 1 The SP500 option market generated a large literature over the last 50 years, spanning several fields in economics and finance;
Asset pricing, market design, microstructure, risk sharing and efficiency, macro-finance...
- 2 ... but short-dated options were largely ignored, considered illiquid, only worth a footnote;
Bakshi, Cao and Chen (1997) used days-to-maturity < 6 to avoid “liquidity-related biases”. The filter became standard.
- 3 ... now 0DTEs generate more than half of the trading volume in the SP500 option market.
Renewed attention: several timely and complementary papers on 0DTEs.

Focus: Option-Implied U-Shaped Pricing Kernel



Intuitive merge: intermediation frictions are short-lived.

This Paper

Ambitious: the cross-section of 0DTE *intraday* returns, the implied risk-neutral distribution, and the implied pricing kernel.

- **Take-Aways:**

- ❶ Large negative returns for 0DTE put *and* call options (fact),
- ❷ strongly U-shaped intraday pricing kernel (estimated).
- ❸ Bonus: the 1-month kernel is monotonous (no more U-shape at this horizon).

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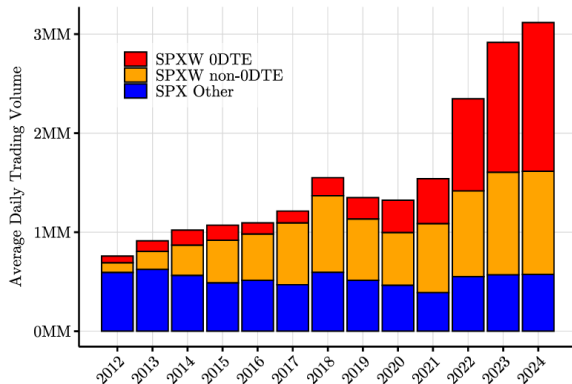
Fascinating and carefully executed empirical results. Timely and relevant. Well written.

A big deal: this may help understand why this contract has become so important.

Comments

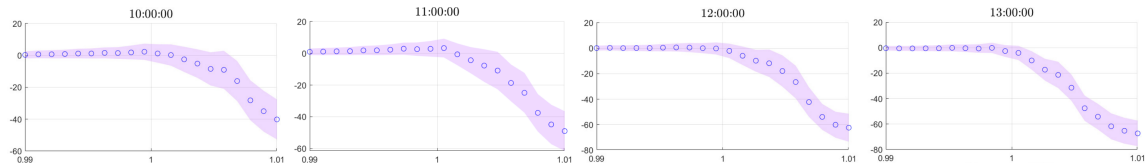
- 1 The horizon gradient — is 0DTE a regime, or the short end of a continuum?
- 2 Trading opportunities — what is the risk-returns tradeoff?
- 3 Who bears the risk — market-makers, or end-users?
- 4 When the book is built — during 0DTE trading, or days before?

0DTE Trading Volume Surges from 2022



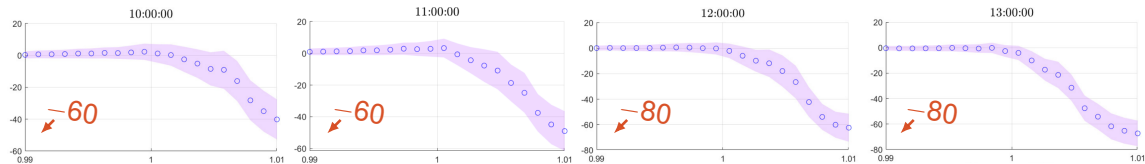
But note that the SPXW volumes surge too, for contracts that will *become* 0DTEs in a *few days*.

Comment 1: 0DTE Call Options Hold-to-Maturity Returns



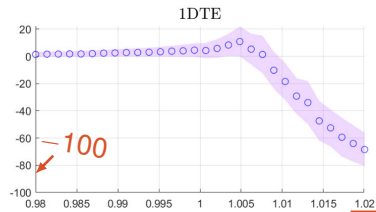
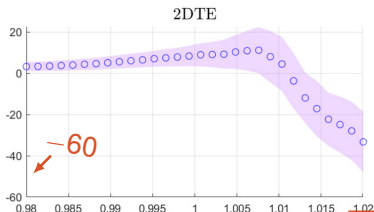
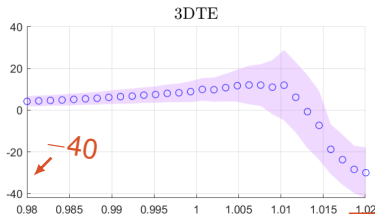
- Result 1: OTM 0DTE Calls are expensive relative to realized index returns — writing naked Calls appears profitable.
- Result 2: The negative slope strongly points to a positive slope for the pricing kernel over this interval — the logic of Bakshi, Madan and Panayotov (2010).
- (Puts are expensive too, but presumably less of a puzzle?)

Comment 1: 0DTE Call Options Hold-to-Maturity Returns



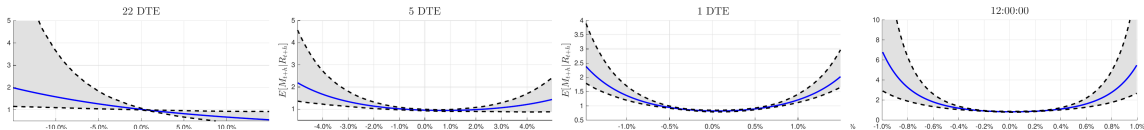
- Result 3? The curve is “steeper” in the later hours of the day
- The y-axis here stretches from -60 to -80 , so the drawn slopes hide how the actual gradient steepens. The paper reads the panels as “similar across different times of the day”.

Comment 1: SPXW Call Options Hold-to-Maturity Returns



- Result 3, again? The right tail also deepens in the days heading into the 0DTE day.
- The moneyness range on the x -axis is wider than in the intraday results above, hiding that the earlier intraday slope was steeper.
- OTM puts offer the opposite but intuitive pattern: they become relatively *more* expensive at *longer* horizons.

Comment 1: This Maps into the Estimated Pricing Kernels



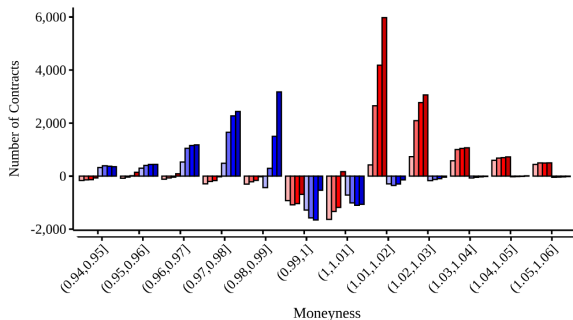
- Same pattern in estimated kernels: monotone at the 1-month horizon, and progressively more U-shaped as maturity shortens.
- Q: Why are progressively rarer events carrying progressively more expensive options as time goes by? A: End-users value leverage? $\Omega = \frac{S\Delta}{C} \approx \frac{1}{\sigma\sqrt{\tau}}$

Comment 2: If They Get Expensive, Why Not Buy Them Early?

- A predictable pattern with rising prices peaking during the maturity day — trade it?
 - ▶ Buy calls in the morning, or days before, and sell them during the 0DTE afternoon.
- But the pattern survives. Because carry and risks prevent trading from closing the gap?
 - ▶ It seems costly, risky and sophisticated to manage an inventory that will land right in the $[0.99, 1.01]$ moneyness range on the 0DTE afternoon.

Comment 3: Net Open Interest and Hedging Needs

(a) Net Open Interest of *Customers* in 0DTE

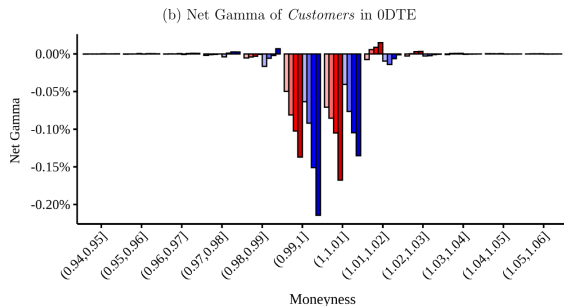


Calls in red, puts in blue — four bars each, at 10:00, 12:00, 14:00 and 15:50, darkening through the day.

- **Net open interest** counts what was built before and during the expiration day:

- ▶ End-users start the day **short** in the ATM strikes $\sim (0.99, 1.01]$.
- ▶ So, *market-makers* may be trading the carry?

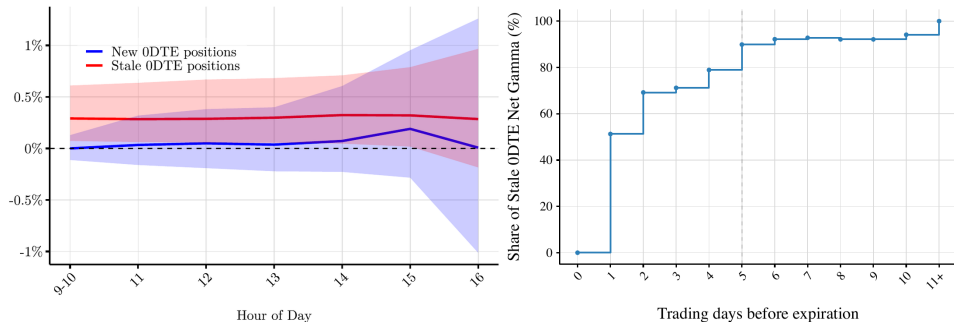
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Calls in red, puts in blue — four bars each, at 10:00, 12:00, 14:00 and 15:50, darkening through the day.

- Market-makers are long **Gamma** (ATM options much more sensitive).
 - ▶ Arguably a “good” inventory: profitable carry + dynamic hedging would require buying the index low and sell it high.

Comment 4: Where the 0DTE Book Comes From



- Stale positions sit near 0.3% of index capitalisation; newly opened 0DTEs close to zero, on average.
- **Built in the days before.** Half of the expiring gamma was opened *one* trading day earlier; about 90% within five.

Conclusion

I like! Read it if you care about 0DTEs.

Pivot?

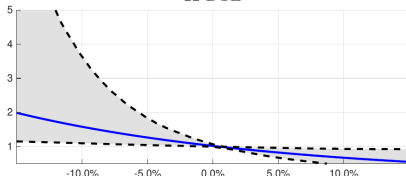
- Focus on the gradual richening as the horizon shortens, starting even a few days before maturity.
- Look at how market-makers 0DTE book is built up over the same few days. Two gradients on the same clock?
- What mechanism could explain how the U-shape has moved to shorter horizons?
 - ▶ End-users are short volatility and this could make calls more expensive (Almeida and Freire 2026, *JF*)?
Or:
 - ▶ End-users value the cheaper leverage of 0DTE contracts?

more stuff that I won't have time to talk about...

Is the 1-Month U-Shape Gone?

What does the average 1-month kernel look like? It seems to be monotone *here*.

22 DTE



	Sample	Average 1-month kernel
Li & Ou	1996–2021	U-shaped
Driessen et al.	1996–2021	U-shaped, drives longer horizon
Almeida & Freire	1996–2020	U-shaped on average <i>but</i> can flip
This paper	2012–2025	monotone decreasing

- Every day in this sample is a **0DTE expiration day** — 1 per week before 2016, progressively more, and all weekdays after May 2022.
- 40% of this sample is data after May 2022, but still a large overlap between the samples.

Is the 1-Month U-Shape Gone?

Two potential sources of explanations — and each has its own test:

- **(i) Selecting days with 0DTEs.** The sample *selects* 0DTE expiration days, so the counterfactual day is absent by construction.
 - ▶ Test: put the non-0DTE days back. Are the average returns and the estimated kernels different on those days?
- **(ii) The volume of 0DTE.** A structural break in 2022, with the post-surge data dominating the full-sample estimates.
 - ▶ Test: split at 11 May 2022. By the way, you already split these other objects in the appendix — the variance risk premium, predictability, the bounds, the strategies. Not the kernel.

Not the same data either: OptionMetrics end-of-day quotes for the first three, Cboe 1-minute quotes here. Hard to see that moving a result this far — but worth remembering.

Equity risk premium

Maybe discuss the peak above 1, also starting to appear around 3DTEs.

Variance risk premium

Maybe discuss the same, appearing around 3DTEs.

Predictability

Is the evidence that VRP+ forecasts realized returns with a negative sign really consistent with U-shaped? Or is this additional evidence of segmentation?